Graduate Research Progress System — Product Requirements Document

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1. Executive summary

The Graduate Research Progress System (GRPS) is a web application, built on a Spring Boot backend, that manages every Masters and PhD student from enrolment to an approved thesis across a two-year, four-semester programme. It replaces spreadsheets, paper panel sheets and email chains with one system that enforces the school's progression rules automatically.

Document version 1.0. It supersedes the earlier project overview and is the baseline for design, development and testing.

The system is built around six rules:

- **Sequential presentations.** Masters students give 5 research presentations (the 5th is the final pre-defense); PhD students give 6. Each is scheduled by the admin, recorded by the secretary, and must be passed and admin-approved before the next unlocks.
- **Time limits.** Both programmes last 2 years (4 semesters). Students need 4 approved presentations by the 18-month mark; anyone who has not completed all presentations by the end of semester 4 is flagged for expulsion.
- **Verified publications.** Students submit publications for librarian review; a publication counts only after the librarian approves it.
- **Screened theses.** Every thesis is checked automatically for format, then for similarity and AI-generated content; a score above 20% is rejected.
- **Attendance-based exam eligibility.** Lecturers record attendance; a student below 70% in a course cannot sit that course's exam.
- **Mandatory supervision.** Every student has a supervisor, who is always a lecturer.

2. Goals and success metrics

The system succeeds when every progression decision is made in it, by rule, and every student can see where they stand without asking anyone. Targets below are proposed launch targets for the first full academic year.

Goal	Metric	Target
One source of truth for student progress	Share of presentations scheduled, recorded and approved in the system	100%
Rules enforced, not remembered	Presentations scheduled out of sequence	0
Faster approvals	Median time from outcome recorded to admin approval	Under 5 working days
Students always informed	Share of state changes that trigger a notification within 1 minute	100%
Early warning on deadlines	Students reaching the 18-month checkpoint without a prior at-risk warning	0
Clean theses	Theses reaching examiners that failed format or screening	0
Fair exam eligibility	Exam lists generated from recorded attendance	100% of courses
Reliable platform	Monthly availability during term	99.5%

Non-goals for version 1: replacing the student records / registry system, fee management, course registration, and online examination delivery.

3. Scope

Version 1 covers the full research progression cycle plus lecture attendance; anything that belongs to the registry, finance or examinations office stays outside.

In scope (v1)

- User accounts, multi-role access (a lecturer can also be a supervisor), programmes, cohorts, semesters
- Supervisor assignment for every student
- Presentation milestone templates (Masters 5, PhD 6), scheduling, outcome recording, sequential gating
- Evidence upload and viewing; admin verification and approval
- Student progress dashboard and achievements timeline
- Publication submission and librarian review
- Thesis submission, automated format check, similarity and AI-content screening

- Lecturer attendance recording and exam eligibility (70% threshold)
- 18-month checkpoint, at-risk warnings, expulsion flagging and admin confirmation
- In-app and email notifications
- Admin settings, reports and audit trail
- Responsive web UI with a persistent left navigation

Out of scope (v1)

- Native mobile apps (the web UI is responsive instead)
- Integration with the student records system (data is imported by CSV in v1)
- Examiner workflow after thesis acceptance (defense scheduling, examiner reports, grading)
- Fees, registration and transcripts
- SMS notifications (designed for, not built)
- Single sign-on with the university directory (designed for, not built)

4. Glossary

Term	Meaning in this document
Milestone	One required research presentation in a student's programme, identified by its sequence number (1–5 for Masters, 1–6 for PhD)
Presentation attempt	One sitting of a milestone; a failed milestone gets a new attempt
Final pre-defense	The last milestone; passing it makes the student eligible to submit the thesis
Outcome	The panel decision recorded by the secretary: PASSED, FAILED, or NOT_HELD
Approval	The admin's verification that a passed milestone and its evidence are complete; approval unlocks the next milestone
Evidence	Files proving a presentation took place: slides, written report, signed panel sheet, photos
Checkpoint	The 18-month mark, by which 4 milestones must be approved

Term	Meaning in this document
Programme end	The end of semester 4 (24 months after enrolment), by which all milestones must be approved
At risk	Student status set when a checkpoint or programme end is approaching or missed
Flagged for expulsion	Status set automatically at programme end if any milestone is incomplete; becomes EXPELLED only after admin confirmation
Similarity index	The percentage of thesis text matching other sources, from the screening provider
AI score	The percentage of thesis text estimated to be AI-generated, from the screening provider
Exam eligibility	Whether a student may sit a course's exam, based on attendance of at least 70%
Supervisor	A lecturer assigned to guide a student; the Supervisor role is added to a lecturer account

5. Users, roles and permissions

Seven roles; one account can hold several, and the UI shows a role switcher when it does (a lecturer who supervises sees both Lecturer and Supervisor views).

Role	Who	Responsibilities
Super Admin	IT / system owner	Creates admin accounts, manages system configuration, views audit logs
Admin	Graduate school officer	Manages programmes, cohorts, students and staff; assigns supervisors; schedules presentations; verifies and approves milestones; sets thresholds; confirms expulsions
Secretary	Graduate school secretary	Records presentation outcomes and panel remarks; marks presentations as held; keeps student progress current

Role	Who	Responsibilities
Student	Masters or PhD candidate	Views schedule and progress; uploads evidence; submits publications and thesis; views achievements, attendance and eligibility
Lecturer	Teaching staff	Creates class sessions and records attendance for assigned courses
Supervisor	A lecturer with the Supervisor role	Monitors supervisees, comments on evidence and thesis drafts, receives their alerts
Librarian	Library staff	Reviews and approves or rejects publications

Permission matrix (C = create, R = read, U = update, A = approve / decide, own = own records only, sup = supervisees only)

Capability	Admin	Secretary	Student	Lecturer	Supervisor	Librarian
Manage users, programmes, cohorts	CRU	R	—	—	—	—
Assign supervisor / grant roles	CU	—	—	—	—	—
Schedule presentation	CU	R	R own	—	R sup	—
Record outcome	R	CU	R own	—	R sup	—
Upload evidence	R	R	CU own	—	R sup	—
Approve / return milestone	A	R	R own	—	R sup	—
Submit publication	R	—	CU own	—	R sup	R
Review publication	R	—	—	—	—	A
Submit thesis	R	—	CU own	—	R sup	—
Override thesis decision	A	—	—	—	—	—
Record attendance	R	—	—	CU own courses	—	—
View attendance / eligibility	R	R	R own	R own courses	R sup	—
Comment on evidence / thesis	C	—	R own	—	C sup	—
Confirm expulsion	A	—	—	—	R sup	—
Change settings	U	—	—	—	—	—
View reports	R	R	—	R own courses	R sup	R publications

Super Admin has every Admin permission plus admin account management and the audit log.

6. Programme structure

Both programmes run 24 months over 4 semesters; they differ only in the number of presentations. All values below are stored per programme and editable by the admin.

Parameter	Masters	PhD
Duration	24 months	24 months
Semesters	4	4
Presentations required	5	6
Final milestone	Presentation 5 = final pre-defense	Presentation 6 = final pre-defense

Parameter	Masters	PhD
First presentation possible from	Semester 2	Semester 2
18-month checkpoint	4 presentations approved	4 presentations approved
Programme end (all approved)	End of semester 4	End of semester 4
Evidence upload	Mandatory before approval	Optional (can be made mandatory)
Publications	Recorded; count set by programme rules	Recorded; count set by programme rules

Key dates per student (computed from the enrolment date, recalculated if an admin records an approved suspension or extension)

When	What the system does
Month 0	Student enrolled; milestones generated as LOCKED; supervisor must be assigned
Start of semester 2	Milestone 1 becomes AVAILABLE for scheduling
Months 12, 15, 17	Checkpoint warnings if fewer than 4 milestones are approved
Month 18	Checkpoint evaluated; student set to AT_RISK if fewer than 4 approved
Months 18, 21, 23	Programme-end warnings if any milestone is outstanding
Month 24 (end of semester 4)	Student set to FLAGGED_FOR_EXPULSION if any milestone is outstanding; admin reviews
After final pre-defense approved	Thesis submission unlocked

7. Functional requirements — identity, users and programmes

Requirements in sections 7–15 use MoSCoW priority (Must, Should, Could). Every Must requirement is in the v1 release.

ID	Requirement	Priority	Acceptance criteria
FR-AUTH-01	Users sign in with email or staff / student number and password	Must	Valid credentials return an access token (15 min) and refresh token (7 days); invalid ones return 401 with no hint which field was wrong
FR-AUTH-02	Accounts lock after repeated failed sign-ins	Must	5 failures in 15 minutes lock the account for 15 minutes; admin can unlock
FR-AUTH-03	Users reset a forgotten password by email	Must	Reset link is single-use and expires after 30 minutes
FR-AUTH-04	New accounts must set a password at first sign-in	Must	Admin-created accounts receive an activation email; temporary passwords are never emailed
FR-AUTH-05	Users with several roles switch role without signing out	Must	Role switcher appears only when the account holds 2+ roles; the sidebar changes to that role's items
FR-USR-01	Admin creates, edits, deactivates users	Must	Deactivated users cannot sign in; their history is kept
FR-USR-02	Admin bulk-imports students and staff from CSV	Must	Import validates every row first and reports errors per row; nothing is saved if any row fails
FR-USR-03	Admin grants or removes roles on an account	Must	Supervisor role can only be granted to an account that already has the Lecturer role
FR-USR-04	Admin assigns a supervisor to each student	Must	A student cannot be activated without a supervisor; supervisor must hold Lecturer + Supervisor roles; change history is kept
FR-USR-05	Admin can assign an optional co-supervisor	Could	Co-supervisor gets the same read access and notifications
FR-PRG-01	Admin manages programmes (Masters, PhD) with duration, semesters, milestone count, checkpoint rules	Must	Changing a programme does not alter milestones already generated for enrolled students

ID	Requirement	Priority	Acceptance criteria
FR-PRG-02	Admin manages milestone templates per programme	Must	Each template has sequence number, title, description, evidence-required flag, final-pre-defense flag; exactly one final pre-defense per programme
FR-PRG-03	Admin manages academic years, semesters and their dates	Must	Semesters cannot overlap; current semester is derived from today's date
FR-PRG-04	Admin manages cohorts (intake year and semester)	Must	Students belong to exactly one cohort
FR-PRG-05	Enrolling a student generates their milestones	Must	Milestones are created from the programme template at enrolment, all LOCKED, with milestone 1 opening at semester 2
FR-PRG-06	Admin records an approved suspension or extension	Should	Deadlines shift by the recorded number of months; the reason and approver are stored

8. Functional requirements — presentation milestones and scheduling

The admin schedules every presentation, the secretary records what happened, and the system refuses any step taken out of order.

ID	Requirement	Priority	Acceptance criteria
FR-MIL-01	Each student has 5 (Masters) or 6 (PhD) milestones in fixed order	Must	Milestone list shows sequence, title, status, scheduled date, outcome and approval date
FR-MIL-02	Admin schedules a milestone: date, start time, duration, venue or online link, panel members	Must	Allowed only when the milestone is AVAILABLE (previous one APPROVED); otherwise the API returns 409 `MILESTONE_LOCKED` with the blocking milestone

ID	Requirement	Priority	Acceptance criteria
FR-MIL-03	Admin can schedule many students into one presentation session	Should	Bulk scheduling assigns time slots in order; each student still gets an individual record
FR-MIL-04	System prevents clashes	Must	A student cannot have two presentations at overlapping times; a venue cannot be double-booked; a panel member clash shows a warning
FR-MIL-05	Admin reschedules or cancels a scheduled presentation	Must	Reason is mandatory; student, supervisor and secretary are notified; history is kept
FR-MIL-06	Secretary sees a daily and weekly list of scheduled presentations	Must	Filterable by date, programme, venue, status
FR-MIL-07	Secretary records the outcome: PASSED, FAILED or NOT_HELD, with panel remarks and optional score	Must	Outcome can only be recorded on or after the scheduled date; remarks are required for FAILED
FR-MIL-08	A FAILED or NOT_HELD milestone returns to AVAILABLE for a new attempt	Must	A new attempt record is created; previous attempts stay visible
FR-MIL-09	Maximum attempts per milestone are configurable	Should	Default 3; exceeding it notifies the admin and sets the student AT_RISK
FR-MIL-10	A PASSED milestone moves to AWAITING_VERIFICATION	Must	Admin is notified; for Masters the student is prompted to upload evidence
FR-MIL-11	Admin approves or returns the milestone (see section 9)	Must	Approval sets APPROVED and makes the next milestone AVAILABLE in the same transaction
FR-MIL-12	Approving the final pre-defense unlocks thesis submission	Must	Student sees the Thesis page become active and gets a notification
FR-MIL-13	Secretary can correct an outcome within 48 hours	Should	After 48 hours only the admin can correct it; every change is audited

ID	Requirement	Priority	Acceptance criteria
FR-MIL-14	Students and supervisors can export the schedule to a calendar	Could	Each scheduled presentation offers an .ics download

9. Functional requirements — evidence, verification, progress and achievements

A passed presentation counts only after the admin has checked its evidence and approved it.

ID	Requirement	Priority	Acceptance criteria
FR-EVD-01	Students attach evidence to a milestone: slides, written report, signed panel sheet, photos	Must	Accepted types PDF, DOCX, PPTX, JPG, PNG; max 25 MB per file, 10 files per milestone; files are virus-scanned before they are stored
FR-EVD-02	Masters students must upload at least one evidence file before approval	Must	Approve action is disabled and the API returns 422 `EVIDENCE_REQUIRED` until a file exists
FR-EVD-03	Students, their supervisor and admins can view and download evidence	Must	Preview in browser for PDF and images; downloads use signed links valid for 10 minutes
FR-EVD-04	Students can replace evidence until the milestone is approved	Must	Replaced files are kept as earlier versions; approved evidence is read-only
FR-VER-01	Admin sees a verification queue of milestones AWAITING_VERIFICATION	Must	Sorted oldest first; shows student, programme, milestone, outcome date, evidence count, days waiting
FR-VER-02	Admin approves a milestone	Must	Sets APPROVED, records approver and time, unlocks the next milestone, notifies student and supervisor

ID	Requirement	Priority	Acceptance criteria
FR-VER-03	Admin returns a milestone with comments	Must	Comment is mandatory; status becomes RETURNED; student is notified and can upload again, which moves it back to AWAITING_VERIFICATION
FR-VER-04	Supervisor can comment on evidence before admin review	Should	Comments are visible to student and admin
FR-DSH-01	Student dashboard shows progress at a glance	Must	Shows programme, semester (1–4), milestones approved out of required, next action, days to checkpoint and to programme end, status (On track / At risk / Flagged)
FR-DSH-02	Locked milestones explain why	Must	E.g. "Locked until Presentation 2 is approved"
FR-ACH-01	Achievements page lists every approved accomplishment in date order	Must	Approved milestones, librarian-approved publications, thesis acceptance; each with date and who approved it
FR-ACH-02	Achievement badges for milestones	Could	Badge per approved milestone and for first approved publication
FR-ACH-03	Student downloads a progress statement as PDF	Should	Statement includes all approved items, generated date and a verification code

10. Functional requirements — publications and librarian review

A publication belongs to a student's record only after a librarian approves it; until then it is a private submission.

ID	Requirement	Priority	Acceptance criteria
FR-PUB-01	Student creates a publication record	Must	Fields: title, type (journal article, conference paper, book chapter, other), journal / conference name, authors, year, volume / issue / pages, DOI or URL, ISSN / ISBN, publication status (accepted, published), full-text PDF (max 25 MB)
FR-PUB-02	Student saves drafts and requests review when ready	Must	DRAFT records are visible only to the student; Submit moves it to SUBMITTED and notifies the librarian queue
FR-PUB-03	System warns about duplicates	Should	A DOI already linked to any student, or a near-identical title, shows a warning to student and librarian
FR-PUB-04	Librarian sees a review queue	Must	Oldest first; filter by programme, type, status; shows days waiting
FR-PUB-05	Librarian opens a submission and verifies it	Must	Sees all metadata, PDF preview, and a DOI link; can mark checklist items (venue verified, authorship verified, indexing checked)
FR-PUB-06	Librarian approves a publication	Must	Status APPROVED; the publication is linked to the student, appears in achievements and reports; student and supervisor are notified
FR-PUB-07	Librarian rejects or requests changes, with a reason	Must	Reason is mandatory; REJECTED is final, CHANGES_REQUESTED lets the student edit and resubmit
FR-PUB-08	Approved publications are read-only	Must	Only an admin can revoke an approval, with a reason, which is audited and notified
FR-PUB-09	Supervisor sees supervisees' submitted and approved publications	Must	Read-only
FR-PUB-10	Look up metadata from a DOI	Could	Entering a DOI pre-fills title, authors, venue and year from Crossref

11. Functional requirements — thesis submission and screening

Every thesis passes two automatic gates in order — format, then similarity / AI screening — before anyone else sees it. A score above 20% is rejected.

ID	Requirement	Priority	Acceptance criteria
FR-THS-01	Thesis submission opens only after the final pre-defense is approved	Must	Thesis page is locked with an explanation until then; API returns 409 `THESIS_LOCKED`
FR-THS-02	Student uploads the thesis	Must	DOCX (preferred, enables the full format check) or PDF; max 50 MB; title, abstract and keywords required; each upload creates a new version
FR-THS-03	System runs the format check automatically on upload	Must	Result within 2 minutes for a 300-page document; status FORMAT_CHECKING then FORMAT_PASSED or FORMAT_FAILED
FR-THS-04	Format check report lists every rule with pass / fail and location	Must	E.g. "Left margin 2.5 cm, required 3.8 cm — pages 12–40"; student can download the report
FR-THS-05	Format rules are configurable by the admin	Must	Rule set per programme; see default rule set below
FR-THS-06	Passing the format check starts similarity and AI screening automatically	Must	Status SCREENING; submitted to the provider via API; result stored with the provider's report
FR-THS-07	Thesis is rejected automatically if the similarity index or AI score exceeds the limit	Must	Limit default 20% for each; score > 20% sets SCREENING_REJECTED; score ⩽ 20% sets SCREENING_PASSED
FR-THS-08	Student, supervisor and admin see the screening report	Must	Shows similarity %, AI %, the limit applied, and a link to the provider's full report
FR-THS-09	Admin can enter a screening result manually	Must	Used when the provider API is unavailable; admin uploads the provider's PDF report and enters both scores; the same 20% rule applies

ID	Requirement	Priority	Acceptance criteria
FR-THS-10	Admin can override a rejection with a written justification	Should	E.g. matches are the student's own approved publications; override is audited and notified
FR-THS-11	A passed thesis moves to the next phase	Must	Status ACCEPTED_FOR_EXAMINATION; admin is notified; examiner workflow is out of scope for v1
FR-THS-12	Rejected theses can be revised and resubmitted	Must	A new version restarts both gates; all versions and reports are kept
FR-THS-13	Supervisor can comment on each thesis version	Should	Comments visible to student and admin

Default format rule set (to be confirmed against the school's thesis guidelines)

Rule code	Check	Default
FMT-PAGE	Paper size	A4
FMT-MARGIN	Margins	Left 3.8 cm (binding), top / right / bottom 2.5 cm
FMT-FONT	Body font and size	Times New Roman 12 pt
FMT-SPACING	Body line spacing	1.5
FMT-HEAD	Heading styles used for chapter titles	Word Heading 1–3 styles present
FMT-PAGENUM	Page numbering	Roman for preliminary pages, Arabic from Chapter 1
FMT-SECTIONS	Required sections present, in order	Title page, Declaration, Abstract, Acknowledgements, Table of contents, Chapters, References, Appendices
FMT-ABSTRACT	Abstract length	300 words maximum
FMT-TOC	Table of contents present and generated	Yes
FMT-REFS	References section present and non-empty	Yes

PDF uploads can only be checked for page size, margins (approximate), page count and section headings; the report says which checks were skipped.

12. Functional requirements — attendance and exam eligibility

Lecturers record attendance per class session; a student whose attendance in a course falls below 70% is not eligible for that course's exam.

ID	Requirement	Priority	Acceptance criteria
FR-ATT-01	Admin manages courses per semester and assigns lecturers and enrolled students	Must	CSV import supported; a course can have more than one lecturer

ID	Requirement	Priority	Acceptance criteria
FR-ATT-02	Lecturer creates a class session: course, date, start / end time, topic	Must	Sessions in the future cannot have attendance marked
FR-ATT-03	Lecturer marks each enrolled student PRESENT, ABSENT, LATE or EXCUSED	Must	Roster shows all enrolled students defaulting to ABSENT; "Mark all present" button; saves in one request
FR-ATT-04	Lecturer can edit a session's attendance for 7 days	Must	After 7 days only an admin can change it, with a reason; all changes audited
FR-ATT-05	Attendance percentage is calculated per student per course	Must	% = (PRESENT + LATE) ÷ (sessions held − EXCUSED) × 100, rounded to one decimal; recalculated on every save
FR-ATT-06	Exam eligibility follows the threshold	Must	⩾ 70% = ELIGIBLE; < 70% = NOT_ELIGIBLE; threshold editable by admin
FR-ATT-07	Students see attendance and eligibility per course	Must	Shows sessions held, attended, percentage, status, and how many more sessions they can miss while staying eligible
FR-ATT-08	Warnings before a student drops below the threshold	Should	Notification when attendance falls below 75%, and again when it falls below 70%
FR-ATT-09	Admin and lecturer export the exam eligibility list per course	Must	CSV and PDF; lists every student with percentage and status; generated as of a chosen date
FR-ATT-10	Admin can grant an eligibility exception	Should	Requires a reason (e.g. documented medical leave); shown as ELIGIBLE (EXCEPTION) and audited
FR-ATT-11	QR-code self check-in	Could	Lecturer shows a code valid for 10 minutes; students scan it while signed in

13. Functional requirements — notifications

Every event that asks someone to act, or changes a student's standing, produces an in-app notification and an email.

ID	Requirement	Priority	Acceptance criteria
FR-NTF-01	In-app notification centre	Must	Bell icon with unread count on every page; list with read / unread, filter by type, mark all read; each item links to the relevant page
FR-NTF-02	Email for every notification	Must	Sent within 1 minute; branded HTML template with plain-text fallback; failed sends retried 3 times
FR-NTF-03	Live updates without page refresh	Should	Unread count updates in real time over WebSocket or SSE
FR-NTF-04	Users choose email frequency for non-urgent items	Could	Immediate or daily digest; deadline, rejection and expulsion notices are always immediate
FR-NTF-05	Admin sends announcements to a group	Should	Target by role, programme, cohort or individual

Notification catalogue

Code	Trigger	Recipients
N-01	Milestone becomes available	Student, supervisor
N-02	Presentation scheduled	Student, supervisor, secretary
N-03	Presentation rescheduled or cancelled	Student, supervisor, secretary
N-04	Reminder 48 h and 24 h before a presentation	Student, supervisor
N-05	Outcome recorded (passed / failed / not held)	Student, supervisor; admin when passed
N-06	Evidence uploaded	Admin
N-07	Milestone approved	Student, supervisor

Code	Trigger	Recipients
N-08	Milestone returned with comments	Student, supervisor
N-09	Publication submitted for review	Librarians
N-10	Publication approved / changes requested / rejected	Student, supervisor
N-11	Thesis format check passed / failed	Student, supervisor
N-12	Thesis screening passed / rejected	Student, supervisor, admin
N-13	Thesis accepted for examination	Student, supervisor, admin
N-14	Attendance below 75% / below 70% in a course	Student, lecturer
N-15	Checkpoint or programme-end warning	Student, supervisor, admin
N-16	Status changed to AT_RISK	Student, supervisor, admin
N-17	Flagged for expulsion	Student, supervisor, admin
N-18	Expulsion confirmed or flag cleared	Student, supervisor
N-19	Supervisor assigned or changed	Student, old and new supervisor
N-20	Supervisor comment added	Student

14. Functional requirements — deadlines, risk and expulsion

The system watches every student's deadlines nightly, warns early, and flags expulsion automatically — but a person always makes the final expulsion decision.

ID	Requirement	Priority	Acceptance criteria
FR-DLN-01	Each student has a computed checkpoint date (enrolment + 18 months) and programme-end date (end of semester 4)	Must	Both shown on the student, supervisor and admin dashboards; shifted by recorded extensions
FR-DLN-02	Nightly job evaluates every ACTIVE and AT_RISK student	Must	Runs at 01:00 Africa/Accra; idempotent; runs once even with several server instances

ID	Requirement	Priority	Acceptance criteria
FR-DLN-03	Checkpoint warnings	Must	At months 12, 15 and 17, a student with fewer than 4 approved milestones receives N-15
FR-DLN-04	Checkpoint evaluation	Must	On the checkpoint date, fewer than 4 approved milestones sets status AT_RISK and sends N-16
FR-DLN-05	Programme-end warnings	Must	At months 18, 21 and 23, any outstanding milestone triggers N-15
FR-DLN-06	Programme-end evaluation	Must	On the programme-end date, any milestone not APPROVED sets FLAGGED_FOR_EXPULSION and sends N-17
FR-DLN-07	Admin reviews flagged students	Must	Expulsion queue shows each student's milestones, attempts, extensions and supervisor comments
FR-DLN-08	Admin confirms expulsion or clears the flag	Must	Both need a written reason; confirm sets EXPELLED, deactivates student sign-in after 30 days, and keeps the record; clear returns the student to ACTIVE or AT_RISK with a new deadline if an extension is recorded
FR-DLN-09	Expelled students cannot be scheduled, submit work or appear on eligibility lists	Must	Enforced in the service layer
FR-DLN-10	Students who complete every milestone and have an accepted thesis are marked COMPLETED	Must	Completed students are excluded from deadline checks
FR-DLN-11	At-risk report	Must	Admin sees all AT_RISK and FLAGGED students with days remaining, filterable by programme and cohort

15. Functional requirements — reports, audit and settings

Admins get standard reports out of the box, every change is audited, and every threshold in this document is a setting rather than code.

ID	Requirement	Priority	Acceptance criteria
FR-RPT-01	Programme progress report	Must	Per programme / cohort: students by status, milestones approved distribution, average days between milestones
FR-RPT-02	Presentation schedule report	Must	Presentations by date range, venue, outcome
FR-RPT-03	Verification backlog report	Must	Milestones awaiting verification with days waiting
FR-RPT-04	Publications report	Must	Approved publications by student, programme, type, year
FR-RPT-05	Thesis screening report	Must	Submissions, format pass rate, screening pass rate, overrides
FR-RPT-06	Attendance and eligibility report	Must	Per course and per student
FR-RPT-07	Supervisor load report	Should	Supervisees per supervisor with their statuses
FR-RPT-08	All reports export to CSV and PDF	Must	Export reflects the filters applied on screen
FR-AUD-01	Audit trail of every create, update, status change and approval	Must	Stores who, when, what entity, old and new values, IP address; cannot be edited or deleted by any role
FR-AUD-02	Admin views the audit log per student or per entity	Must	Filter by user, entity type, date range
FR-SET-01	Admin edits system settings	Must	Similarity limit (20%), AI-score limit (20%), attendance threshold (70%), warning threshold (75%), max attempts (3), checkpoint months (18), checkpoint milestones (4), programme months (24), reminder hours (48, 24), file size limits
FR-SET-02	Setting changes apply from the time of change	Must	Changes are audited; past decisions are not recalculated

ID	Requirement	Priority	Acceptance criteria
FR-SET-03	Admin manages venues and email templates	Should	Venues have name and capacity; templates support placeholders such as student name and milestone

16. Business rules register

Each rule is enforced in the backend service layer (never only in the UI) and has at least one automated test named after its ID.

ID	Rule	Enforced in	Default
BR-01	Masters students complete 5 presentations; the 5th is the final pre-defense	Programme template	5
BR-02	PhD students complete 6 presentations; the 6th is the final pre-defense	Programme template	6
BR-03	A milestone can be scheduled only when the previous milestone is APPROVED	`MilestoneService.schedule`	—
BR-04	Only the admin schedules presentations (date, time, venue)	Security + service	—
BR-05	Only the secretary (or admin) records outcomes	Security + service	—
BR-06	A passed milestone counts only after admin approval	`VerificationService.approve`	—
BR-07	Masters milestones need at least one evidence file before approval	`VerificationService.approve`	1 file
BR-08	A publication is linked to a student only after librarian approval	`PublicationService.approve`	—

ID	Rule	Enforced in	Default
BR-09	Thesis submission requires the final pre-defense to be APPROVED	`ThesisService.submit`	—
BR-10	A thesis must pass the format check before screening	Thesis pipeline	—
BR-11	A thesis with similarity index or AI score above the limit is rejected	Thesis pipeline	20% each
BR-12	Both programmes run 2 years over 4 semesters	Programme settings	24 months
BR-13	4 milestones must be approved by the 18-month checkpoint, else AT_RISK	Deadline job	4 by month 18
BR-14	All milestones must be approved by the end of semester 4, else FLAGGED_FOR_EXPULSION	Deadline job	Month 24
BR-15	Expulsion becomes final only after admin confirmation with a reason	`StudentService.confirmExpulsion`	—
BR-16	A student below the attendance threshold in a course is not eligible for that exam	`EligibilityService`	70%
BR-17	Every active student has exactly one primary supervisor	`StudentService` validation	—
BR-18	A supervisor must hold the Lecturer role	`UserService.grantRole`	—
BR-19	Failed milestones may be re-attempted up to the maximum	`MilestoneService`	3 attempts
BR-20	Expelled and completed students cannot be scheduled or submit work	All write services	—

18. User interface and navigation

The left navigation is permanent: it never collapses, auto-hides or disappears when the page changes. Only the content area to its right changes.

Layout rules

- Fixed left sidebar, 240 px wide on screens ⩾ 1024 px; a fixed 72 px icon rail with labels on hover between 640 and 1023 px; a fixed bottom tab bar on phones under 640 px. None of these can be dismissed.
- Top bar on every page: page title, role switcher (when the user has 2+ roles), notification bell with unread count, profile menu.
- The active page is highlighted in the sidebar; the sidebar keeps its scroll position across pages.
- Locked items stay visible but greyed out, with a tooltip explaining what unlocks them.
- Every list page has search, filters, sorting, pagination (25 rows default) and CSV export where relevant.
- Destructive or final actions (approve, reject, confirm expulsion) need a confirmation dialog stating the consequence.

Sidebar items and screens by role

Role	Sidebar items	Key screens
Student	Dashboard, My Presentations, Evidence, Publications, Thesis, Attendance, Achievements, Notifications, Profile	Progress dashboard with countdowns; milestone timeline; evidence upload; publication form; thesis upload with format and screening reports; attendance per course; achievements timeline
Supervisor	Dashboard, My Students, Comments, Notifications	Supervisee list with status chips; student detail (milestones, evidence, publications, thesis, attendance)
Lecturer	Dashboard, My Courses, Sessions, Attendance, Eligibility Lists, Notifications	Session list; attendance roster with Mark all present; eligibility list export
Secretary	Dashboard, Presentation Calendar, Record Outcomes, Student Progress, Notifications	Day / week calendar; outcome form; progress lookup

Role	Sidebar items	Key screens
Admin	Dashboard, Students, Staff, Programmes, Scheduling, Verification Queue, Publications, Theses, Courses, At-Risk & Expulsion, Reports, Settings, Audit Log	Scheduling board; verification queue; expulsion review; settings; reports
Librarian	Dashboard, Review Queue, Approved Publications, Notifications	Review screen with PDF preview and checklist

Design standards

- WCAG 2.1 AA: keyboard navigable, visible focus, 4.5:1 text contrast, labelled form fields.
- Status colours used consistently: green On track / Approved, amber At risk / Awaiting, red Flagged / Rejected, grey Locked — always with a text label, never colour alone.
- Dates shown as 19 Oct 2026, times in 24-hour Africa/Accra time.
- Supported browsers: latest two versions of Chrome, Edge, Firefox, Safari.

19. Non-functional requirements

Sized for a graduate school of up to 5,000 students and 500 staff, with headroom to grow without redesign.

ID	Area	Requirement
NFR-01	Performance	95% of API reads respond in under 300 ms and writes under 800 ms at 200 concurrent users
NFR-02	Performance	Dashboard pages load in under 2 seconds on a 10 Mbps connection
NFR-03	Performance	Format check of a 300-page DOCX completes in under 2 minutes; runs asynchronously
NFR-04	Scalability	Stateless API instances behind a load balancer; scheduled jobs protected by a distributed lock
NFR-05	Availability	99.5% monthly during term; planned maintenance outside 07:00–20:00 Africa/Accra
NFR-06	Backup	Nightly full database backup and continuous WAL archiving; object storage versioned; restore tested each semester
NFR-07	Recovery	RPO 15 minutes, RTO 4 hours

ID	Area	Requirement
NFR-08	Data retention	Student records, evidence and theses kept for at least 7 years after completion or exit; audit log kept 7 years
NFR-09	Usability	A new student completes evidence upload without help in under 3 minutes (usability test)
NFR-10	Accessibility	WCAG 2.1 AA
NFR-11	Observability	Structured JSON logs with request IDs; health, metrics and readiness endpoints; alerts on error rate and job failures
NFR-12	Maintainability	Backend test coverage ⩾ 80% on service classes; every business rule has a named test; OpenAPI spec generated from code
NFR-13	Portability	Runs as Docker containers on any Linux host or Kubernetes; no cloud-vendor lock-in
NFR-14	Time	All timestamps stored in UTC; displayed in Africa/Accra
NFR-15	Localisation	English UI; strings externalised for future translation

20. Security and data protection

The system holds personal data and academic decisions, so access is least-privilege by default and every decision is traceable.

ID	Requirement
SEC-01	All traffic over HTTPS (TLS 1.2+); HSTS enabled
SEC-02	Passwords hashed with BCrypt (cost 12); minimum 10 characters with letters and numbers
SEC-03	JWT access tokens signed with RS256, 15-minute lifetime; refresh tokens 7 days, stored hashed, rotated on use, revocable

ID	Requirement
SEC-04	Role-based access enforced on every endpoint with `@PreAuthorize`; ownership checks (own records, own courses, supervisees) enforced in services
SEC-05	Object-level checks on every ID in a URL — a student requesting another student's evidence gets 404
SEC-06	Uploaded files: type checked by content, not extension; virus-scanned with ClamAV; stored outside the web root; served only through short-lived signed URLs
SEC-07	Input validation with Bean Validation on every DTO; parameterised queries only (JPA)
SEC-08	Rate limiting: 10 sign-in attempts per minute per IP; 100 requests per minute per user on write endpoints
SEC-09	CORS restricted to the frontend origin; CSRF not needed for bearer-token APIs; secure headers (CSP, X-Frame-Options) on the frontend
SEC-10	Secrets (DB password, JWT keys, provider API keys) from environment variables or a secrets manager, never in the repository
SEC-11	Dependency and container scanning in CI (OWASP Dependency-Check, Trivy); critical findings block release
SEC-12	Personal data processed in line with Ghana's Data Protection Act, 2012 (Act 843): purpose limitation, access on a need-to-know basis, data subject access requests handled by the admin
SEC-13	Penetration test against the OWASP Top 10 before go-live

22. Data model

PostgreSQL, one schema, UUID primary keys, snake_case names. Every table carries `id`, `created_at`, `updated_at`, `created_by`, `updated_by` and `version` (optimistic locking); these are not repeated below.

Table	Key columns	Constraints and notes
`app_user`	email, staff_or_student_no, first_name, last_name, phone, password_hash, active, locked_until, failed_logins, last_login_at	email and number unique; soft-deactivated, never deleted
`role`	code (SUPER_ADMIN, ADMIN, SECRETARY, STUDENT, LECTURER, SUPERVISOR, LIBRARIAN)	Seeded by migration
`user_role`	user_id, role_id	PK (user_id, role_id)
`refresh_token`	user_id, token_hash, expires_at, revoked_at	Index on token_hash
`programme`	code, name, level (MASTERS, PHD), duration_months, semesters, required_milestones, checkpoint_months, checkpoint_milestones, evidence_required	code unique
`milestone_template`	programme_id, sequence_no, title, description, is_final_pre_defense, evidence_required	Unique (programme_id, sequence_no); one final per programme
`academic_semester`	academic_year, number (1–2), starts_on, ends_on	No overlapping ranges
`cohort`	programme_id, name, intake_semester_id	

Table	Key columns	Constraints and notes
`student`	user_id, programme_id, cohort_id, enrolled_on, checkpoint_due_on, programme_end_on, current_semester (1–4), status (ACTIVE, AT_RISK, FLAGGED_FOR_EXPULSION, EXPELLED, COMPLETED, WITHDRAWN), status_reason	user_id unique; due dates recomputed on extension
`supervision`	student_id, supervisor_user_id, type (PRIMARY, CO), starts_on, ends_on	Exactly one open PRIMARY per active student; supervisor must hold LECTURER + SUPERVISOR
`extension`	student_id, months, reason, approved_by, approved_on	Shifts both due dates
`student_milestone`	student_id, template_id, sequence_no, status (LOCKED, AVAILABLE, SCHEDULED, AWAITING_VERIFICATION, RETURNED, APPROVED), approved_by, approved_at	Unique (student_id, sequence_no)
`presentation_attempt`	student_milestone_id, attempt_no, scheduled_start, scheduled_end, venue_id, online_link, outcome (PASSED, FAILED, NOT_HELD), score, panel_remarks, recorded_by, recorded_at, cancelled_reason	Unique (student_milestone_id, attempt_no)
`panel_member`	attempt_id, user_id or external_name, role (CHAIR, MEMBER)	
`venue`	name, capacity, active	
`evidence_file`	student_milestone_id, storage_key, file_name, content_type, size_bytes, sha256, kind (SLIDES, REPORT, PANEL_SHEET, PHOTO, OTHER), version, superseded_by, scan_status	Read-only once milestone APPROVED

Table	Key columns	Constraints and notes
`review_comment`	target_type (EVIDENCE, MILESTONE, THESIS), target_id, author_id, body	
`publication`	student_id, title, type, venue_name, authors, year, volume, issue, pages, doi, url, issn_isbn, pub_status, storage_key, review_status (DRAFT, SUBMITTED, CHANGES_REQUESTED, APPROVED, REJECTED), reviewed_by, reviewed_at, review_note	Partial unique index on doi where review_status = APPROVED
`thesis_submission`	student_id, version_no, title, abstract, keywords, storage_key, file_type, status (UPLOADED, FORMAT_CHECKING, FORMAT_FAILED, FORMAT_PASSED, SCREENING, SCREENING_REJECTED, SCREENING_PASSED, ACCEPTED_FOR_EXAMINATION), similarity_pct, ai_pct, limit_applied_pct, provider_ref, provider_report_key, override_by, override_reason	Unique (student_id, version_no)
`format_check_result`	thesis_submission_id, rule_code, passed, detail, location	
`format_rule`	programme_id, rule_code, parameters (JSONB), active	
`course`	code, title, semester_id, credit_hours	Unique (code, semester_id)
`course_lecturer`	course_id, lecturer_user_id	
`course_enrolment`	course_id, student_id	Unique pair
`class_session`	course_id, held_on, starts_at, ends_at, topic, created_by	

Table	Key columns	Constraints and notes
`attendance_record`	session_id, student_id, status (PRESENT, ABSENT, LATE, EXCUSED), marked_by, marked_at	Unique (session_id, student_id)
`eligibility`	course_id, student_id, sessions_held, sessions_counted, attendance_pct, status (ELIGIBLE, NOT_ELIGIBLE, EXCEPTION), exception_reason	Unique pair; recalculated on save
`notification`	recipient_id, code, title, body, link, read_at, email_status, email_attempts	Index on (recipient_id, read_at)
`system_setting`	key, value, value_type, description	key unique
`audit_log`	actor_id, action, entity_type, entity_id, old_value (JSONB), new_value (JSONB), ip_address, at	Append-only; no update or delete grants
`shedlock`	name, lock_until, locked_at, locked_by	Used by ShedLock

23. API specification

REST over JSON at `/api/v1`, bearer-token authenticated, documented by springdoc-openapi at `/swagger-ui.html`. The OpenAPI file generated from code is the contract the frontend builds against.

Conventions

- Resource names plural, kebab-case; IDs are UUIDs.
- Lists return `{ "content": [...], "page": 0, "size": 25, "totalElements": 132, "totalPages": 6 }`; query params `page`, `size`, `sort=field,asc`.
- Dates ISO 8601 (`2026-10-19`), timestamps ISO 8601 UTC (`2026-10-19T09:30:00Z`).
- Status changes are explicit action endpoints (`POST .../approve`), not generic PATCHes of a status field.
- Uploads are `multipart/form-data`; downloads return a signed URL valid for 10 minutes.
- Optimistic locking: updates send `version`; a stale version returns 409 `CONCURRENT_UPDATE`.

Error format (RFC 7807 problem details)

```
{
  "type": "https://grps/errors/milestone-locked",
  "title": "Milestone is locked",
  "status": 409,
  "code": "MILESTONE_LOCKED",
  "detail": "Presentation 3 cannot be scheduled until Presentation 2 is
approved.",
  "instance": "/api/v1/admin/students/8f1c.../milestones/3/schedule",
  "traceId": "a1b2c3d4"
}
```

HTTP	Code	When
400	VALIDATION_FAILED	Bean validation errors; body includes `errors[]` per field
401	UNAUTHENTICATED	Missing or expired token
403	FORBIDDEN	Role not allowed
404	NOT_FOUND	Missing, or not visible to this user
409	MILESTONE_LOCKED, THESIS_LOCKED, SCHEDULE_CLASH, CONCURRENT_UPDATE, INVALID_STATE	Rule or state conflict
422	EVIDENCE_REQUIRED, SUPERVISOR_REQUIRED, MAX_ATTEMPTS_REACHED	Business rule not met
429	RATE_LIMITED	Too many requests

Endpoint catalogue

Method	Path	Role	Purpose
POST	`/auth/login`	Public	Sign in; returns access + refresh tokens and roles
POST	`/auth/refresh`	Public	Rotate refresh token
POST	`/auth/logout`	Any	Revoke refresh token
POST	`/auth/password/forgot`, `/auth/password/reset`	Public	Password reset

Method	Path	Role	Purpose
GET	`/me`	Any	Profile, roles, active role
GET/POST/PUT	`/admin/users`, `/admin/users/{id}`	Admin	List, create, edit users
PUT	`/admin/users/{id}/roles`	Admin	Set roles (Supervisor requires Lecturer)
POST	`/admin/users/import`	Admin	CSV import (validate-only flag supported)
GET/POST/PUT	`/admin/programmes`, `/admin/programmes/{id}/milestone-templates`	Admin	Programmes and templates
GET/POST	`/admin/semesters`, `/admin/cohorts`, `/admin/venues`	Admin	Calendar and reference data
POST	`/admin/students`	Admin	Enrol student (generates milestones; supervisor required)
PUT	`/admin/students/{id}/supervisor`	Admin	Assign or change supervisor
POST	`/admin/students/{id}/extensions`	Admin	Record extension
GET	`/students/me/dashboard`	Student	Progress, due dates, next action
GET	`/students/me/milestones`	Student	Milestones with attempts
GET	`/students/me/achievements`	Student	Achievement timeline
GET	`/students/me/statement.pdf`	Student	Progress statement
POST	`/admin/milestones/{id}/schedule`	Admin	Schedule an attempt
POST	`/admin/schedule/bulk`	Admin	Schedule many students into a session
POST	`/admin/attempts/{id}/reschedule`, `/cancel`	Admin	Change or cancel (reason required)
GET	`/secretary/presentations?from=&to=`	Secretary	Calendar list
POST	`/secretary/attempts/{id}/outcome`	Secretary	Record PASSED / FAILED / NOT_HELD
POST	`/milestones/{id}/evidence`	Student	Upload evidence

Method	Path	Role	Purpose
GET	`/milestones/{id}/evidence`	Student, Supervisor, Admin	List evidence
GET	`/evidence/{id}/download`	Student, Supervisor, Admin	Signed URL
GET	`/admin/verification-queue`	Admin	Milestones awaiting verification
POST	`/admin/milestones/{id}/approve`	Admin	Approve and unlock next
POST	`/admin/milestones/{id}/return`	Admin	Return with comment
POST	`/comments`	Supervisor, Admin	Comment on evidence, milestone or thesis
GET	`/supervisor/students`, `/supervisor/students/{id}`	Supervisor	Supervisees and detail
GET/POST/PUT	`/publications`, `/publications/{id}`	Student	Own publications
POST	`/publications/{id}/submit`	Student	Request librarian review
GET	`/librarian/publications/queue`	Librarian	Review queue
POST	`/librarian/publications/{id}/approve`, `/request-changes`, `/reject`	Librarian	Decision
POST	`/thesis`	Student	Upload thesis version
GET	`/thesis/me`, `/thesis/{id}`	Student, Supervisor, Admin	Versions and status
GET	`/thesis/{id}/format-report`, `/screening-report`	Student, Supervisor, Admin	Reports
POST	`/admin/thesis/{id}/manual-screening`	Admin	Enter scores and upload provider report
POST	`/admin/thesis/{id}/override`	Admin	Override rejection with reason
GET/POST	`/admin/courses`, `/admin/courses/{id}/enrolments`	Admin	Courses and enrolments
GET	`/lecturer/courses`	Lecturer	Own courses

Method	Path	Role	Purpose
POST	`/lecturer/courses/{id}/sessions`	Lecturer	Create session
PUT	`/lecturer/sessions/{id}/attendance`	Lecturer	Save roster
GET	`/lecturer/courses/{id}/eligibility`	Lecturer, Admin	Eligibility list (CSV / PDF via `Accept`)
GET	`/students/me/attendance`	Student	Attendance and eligibility
POST	`/admin/eligibility/{id}/exception`	Admin	Grant exception
GET	`/admin/at-risk`, `/admin/expulsion-queue`	Admin	Risk and expulsion lists
POST	`/admin/students/{id}/expulsion/confirm`, `/clear`	Admin	Decide flagged case
GET	`/notifications`, POST `/notifications/{id}/read`, `/notifications/read-all`	Any	Inbox
GET	`/notifications/stream`	Any	SSE stream of new notifications
GET/PUT	`/admin/settings`	Admin	Settings
GET	`/admin/reports/{report}`	Admin, Secretary	Reports (CSV / PDF via `Accept`)
GET	`/admin/audit`	Admin	Audit log

Example — schedule a presentation

```
POST /api/v1/admin/milestones/5b2e.../schedule
Authorization: Bearer <token>
Content-Type: application/json

{
  "start": "2026-11-12T09:00:00Z",
  "durationMinutes": 45,
  "venueId": "c3a1...",
  "onlineLink": null,
  "panel": [{ "userId": "9d0e...", "role": "CHAIR" }, { "externalName":
"Prof. A. Mensah", "role": "MEMBER" }]
}
```

Returns 201 with the attempt, or 409 `MILESTONE_LOCKED` / `SCHEDULE_CLASH`.

24. Integrations and background jobs

Every external service sits behind a Java interface with a stub implementation, so development and tests never depend on a live provider.

Integration	Interface	v1 implementation	Fallback
Similarity and AI screening	`ScreeningProvider` (`submit`, `fetchResult`)	Adapter for the university's licensed provider (e.g. Turnitin / iThenticate API)	Admin enters scores and uploads the provider's report manually (FR-THS-09)
Email	`EmailSender`	Spring Mail over SMTP (university relay or a transactional email service)	Queued in `notification` and retried 3 times
File storage	`FileStore` (`put`, `signedUrl`, `delete`)	S3-compatible API (MinIO on-premise or a cloud bucket)	—
Virus scanning	`VirusScanner`	ClamAV daemon over TCP	Files held as PENDING_SCAN until scanned
DOI lookup (Could)	`DoiResolver`	Crossref REST API	Manual entry
Student records import	CSV upload	Validated batch import	—

Background jobs (Spring `@Scheduled` + ShedLock; times in Africa/Accra)

Job	Schedule	What it does
`DeadlineEvaluationJob`	Daily 01:00	Applies FR-DLN-03 to FR-DLN-06; updates current semester; writes audit entries
`PresentationReminderJob`	Hourly	Sends N-04 for presentations 48 h and 24 h ahead (once each)
`MilestoneUnlockJob`	Daily 00:30	Opens milestone 1 for students entering semester 2

Job	Schedule	What it does
`EligibilityRecalcJob`	Daily 02:00	Full recalculation as a safety net to the on-save update
`ScreeningPollJob`	Every 5 minutes	Fetches pending screening results from the provider
`EmailRetryJob`	Every 5 minutes	Retries failed emails up to 3 times
`TokenCleanupJob`	Daily 03:00	Deletes expired refresh and reset tokens

Asynchronous processing. Thesis format checks and screening submissions run on a bounded `@Async` executor (4 threads) triggered by a `ThesisUploadedEvent`. Domain events (`MilestoneApprovedEvent`, `PublicationApprovedEvent`, etc.) are published with `ApplicationEventPublisher` and handled after commit (`@TransactionalEventListener`) by the notification and audit modules.

25. Testing and acceptance

A feature is done when its acceptance criteria pass as automated tests, the related business rules have named tests, and the feature has been demonstrated to the graduate school.

Level	Tooling	Scope	Gate
Unit	JUnit 5, Mockito, AssertJ	Services, rule checks, calculators (attendance %, due dates, format rules)	≥ 80% line coverage on `service` packages
Integration	Spring Boot Test, Testcontainers (PostgreSQL, MinIO, MailHog)	Repositories, migrations, controllers with security, event listeners	All pass on every pull request
API contract	springdoc OpenAPI diff in CI	Breaking changes to the spec	Breaking change needs approval
End-to-end	Playwright against a seeded environment	Critical journeys (below)	All pass before each release
Performance	k6	200 concurrent users, mixed workload	NFR-01 met

Level	Tooling	Scope	Gate
Security	OWASP ZAP baseline, Dependency-Check, Trivy	OWASP Top 10, vulnerable libraries	No high or critical findings
User acceptance	Scripted scenarios with admin, secretary, librarian, lecturer and student volunteers	Each role's main tasks	Sign-off by the graduate school

Critical journeys for end-to-end tests

1. Admin enrols a Masters student with a supervisor; milestones 1–5 are created LOCKED.
2. Admin tries to schedule milestone 2 before milestone 1 is approved — refused with MILESTONE_LOCKED.
3. Milestone 1: scheduled → outcome FAILED → rescheduled → PASSED → evidence uploaded → returned → re-uploaded → approved; milestone 2 becomes AVAILABLE.
4. Student submits a publication; librarian requests changes; student resubmits; librarian approves; it appears in achievements.
5. Final pre-defense approved → thesis uploaded → format fails → corrected thesis passes format → screening 23% → rejected → revised thesis at 14% → accepted for examination.
6. Lecturer records 10 sessions; a student attending 6 shows 60% and NOT_ELIGIBLE on the exam list.
7. Clock moved to month 18 with 3 approved milestones → student AT_RISK; moved to month 24 with milestones outstanding → FLAGGED_FOR_EXPULSION → admin confirms → EXPELLED.
8. A lecturer granted the Supervisor role switches roles and sees supervisees.

Time-dependent logic uses an injected `java.time.Clock`, so tests can move the date without waiting.

26. Development setup

Everything a developer needs on day one: prerequisites, repository layout, local services, configuration and the first tasks.

Prerequisites

- JDK 21 (Temurin), Maven 3.9+ (or the included Maven wrapper)
- Docker and Docker Compose

- Node.js 20 LTS (frontend)
- IDE: IntelliJ IDEA or VS Code with Java extensions

Generate the backend at start.spring.io: Maven, Java 21, Spring Boot 3 (latest stable), group `edu.university`, artifact `grps-api`, dependencies: Spring Web, Validation, Spring Data JPA, PostgreSQL Driver, Flyway, Spring Security, OAuth2 Resource Server (for JWT), Java Mail Sender, Actuator, Lombok, Spring Boot DevTools, Testcontainers. Then add: springdoc-openapi-starter-webmvc-ui, MapStruct, ShedLock (spring + jdbc-template provider), AWS SDK v2 S3 (or MinIO client), Apache POI (poi-ooxml), Apache PDFBox, OpenPDF (PDF reports).

Repository layout

```
grps/
├── grps-api/                       # Spring Boot backend
│   ├── src/main/java/edu/university/grps/
│   │   ├── GrpsApplication.java
│   │   ├── common/                 # BaseEntity, errors, audit, paging, Clock
config
│   │   ├── security/               # JWT, SecurityConfig, CurrentUser
│   │   ├── user/                   # users, roles, supervision
│   │   ├── programme/              # programmes, templates, semesters,
cohorts
│   │   ├── milestone/              # milestones, attempts, scheduling,
verification
│   │   ├── evidence/
│   │   ├── publication/
│   │   ├── thesis/                 # submission, formatcheck/, screening/
│   │   ├── attendance/             # courses, sessions, eligibility
│   │   ├── deadline/               # deadline evaluation, expulsion
│   │   ├── notification/
│   │   ├── report/
│   │   ├── settings/
│   │   └── storage/                # FileStore, VirusScanner
│   ├── src/main/resources/
│   │   ├── application.yml
│   │   ├── db/migration/           # V1__baseline.sql, V2__seed_roles.sql ...
│   │   └── templates/email/
│   └── src/test/java/...
├── grps-web/                       # frontend
├── docker-compose.yml
└── docs/                           # this PRD, ADRs, API spec export
```

Inside each module: `api/` (controllers, DTOs), `domain/` (entities, enums, events), `service/`, `repository/`. Modules call each other only through `service` interfaces.

Local services — `docker-compose.yml`

```
services:
  postgres:
    image: postgres:16
    environment:
      POSTGRES_DB: grps
      POSTGRES_USER: grps
      POSTGRES_PASSWORD: grps
    ports: ["5432:5432"]
    volumes: [pgdata:/var/lib/postgresql/data]
  minio:
    image: minio/minio
    command: server /data --console-address ":9001"
    environment:
      MINIO_ROOT_USER: grps
      MINIO_ROOT_PASSWORD: grps-secret
    ports: ["9000:9000", "9001:9001"]
  mailhog:
    image: mailhog/mailhog
    ports: ["1025:1025", "8025:8025"]
  clamav:
    image: clamav/clamav
    ports: ["3310:3310"]
volumes:
  pgdata:
```

Configuration — `application.yml` (secrets come from environment variables)

```
spring:
  datasource:
    url: ${DB_URL:jdbc:postgresql://localhost:5432/grps}
    username: ${DB_USER:grps}
    password: ${DB_PASSWORD:grps}
  jpa:
    open-in-view: false
    hibernate.ddl-auto: validate
    properties.hibernate.jdbc.time_zone: UTC
  flyway.enabled: true
  mail:
    host: ${MAIL_HOST:localhost}
    port: ${MAIL_PORT:1025}
  servlet.multipart:
    max-file-size: 50MB
    max-request-size: 260MB
grps:
```

```
  timezone: Africa/Accra
  jwt:
    private-key: ${JWT_PRIVATE_KEY}
    public-key: ${JWT_PUBLIC_KEY}
    access-ttl: 15m
    refresh-ttl: 7d
  storage:
    endpoint: ${S3_ENDPOINT:http://localhost:9000}
    bucket: grps
    access-key: ${S3_ACCESS_KEY:grps}
    secret-key: ${S3_SECRET_KEY:grps-secret}
  screening:
    provider: ${SCREENING_PROVIDER:stub}   # stub | turnitin
  cors.allowed-origins: ${WEB_ORIGIN:http://localhost:5173}
management.endpoints.web.exposure.include: health,info,metrics
```

Profiles: `local` (stubs, seeded demo data), `test` (Testcontainers), `staging`, `prod`.

Engineering conventions

- Git: trunk-based with short-lived branches `feature/FR-MIL-02-schedule`; pull requests need one review and green CI.
- Commits reference requirement IDs (e.g. `FR-MIL-02: reject scheduling when previous not approved`).
- Database changes only through Flyway migrations; never edit an applied migration.
- Entities never leave the service layer; controllers return DTOs mapped with MapStruct.
- Every business rule lives in a service method and has a test named after its BR-ID.
- CI (GitHub Actions or GitLab CI): build, unit + integration tests, coverage, OpenAPI diff, Dependency-Check, Trivy, Docker image push.
- Architecture decisions recorded as short ADRs in `docs/adr/`.

First-week checklist

- [ ] Create the repository, generate the Spring Boot project, add the dependencies above
- [ ] Add `docker-compose.yml` and confirm all four services start
- [ ] Write `V1__baseline.sql` for users, roles, programmes, templates, semesters, cohorts, students, supervision, settings, audit
- [ ] Seed roles, Masters and PhD programmes with their 5 and 6 templates, default settings
- [ ] Implement `BaseEntity`, global exception handler (problem details), audit listener, injected `Clock`
- [ ] Implement security: JWT issue / refresh / logout, `SecurityConfig`, `@PreAuthorize` roles, `CurrentUser`
- [ ] Expose Swagger UI and confirm the first endpoint (`GET /me`) works end to end

- [ ] Set up CI with tests and Testcontainers
- [ ] Scaffold the frontend shell with the persistent sidebar and role switcher

27. Delivery plan

Ten two-week sprints (about 20 weeks) to a production release, ordered so the presentation workflow is usable by sprint 4. The plan assumes a team of two backend developers, one frontend developer and a part-time tester; dates are set once the start date is fixed.

Sprint	Weeks	Backend scope	Frontend scope	Demo at end of sprint
0	1–2	Repository, CI, docker-compose, baseline migration, security and JWT, error handling, audit, `GET /me`	App shell, persistent sidebar, sign-in, role switcher	Users sign in and see their role's empty sidebar
1	3–4	Users, roles, CSV import, programmes, templates, semesters, cohorts, enrolment, supervisor assignment	Admin user, programme and student screens	Admin enrols a student with a supervisor; milestones generated
2	5–6	Scheduling, clash checks, outcomes, attempts, sequential gating, reminders	Scheduling board, secretary calendar and outcome form	Admin schedules; secretary records pass / fail
3	7–8	Evidence storage, virus scan, verification queue, approve / return, notifications (in-app + email)	Evidence upload, verification queue, notification centre	Full milestone cycle through to next milestone unlocked
4	9–10	Student dashboard, achievements, progress statement PDF, supervisor views and comments	Student dashboard, achievements, supervisor screens	Student and supervisor see live progress

Sprint	Weeks	Backend scope	Frontend scope	Demo at end of sprint
5	11–12	Publications and librarian review, duplicate checks	Publication form, librarian queue	Publication approved and shown in achievements
6	13–14	Courses, sessions, attendance, eligibility, exceptions, exports	Lecturer roster, student attendance, eligibility lists	Exam eligibility list generated
7	15–16	Thesis upload, format checker (POI / PDFBox), configurable rules, screening provider adapter + manual entry, override	Thesis upload and reports	Thesis passes and fails both gates
8	17–18	Deadline job, at-risk and expulsion workflow, reports, settings UI, audit viewer	At-risk, expulsion queue, reports, settings	Simulated 24-month run ends in a confirmed expulsion
9	19–20	Performance and security testing, fixes, data migration rehearsal, production deployment, runbook	Accessibility fixes, polish	User acceptance sign-off and go-live

Release gate for go-live: all Must requirements pass, critical journeys pass end to end, no high or critical security findings, UAT signed off, backups restored successfully once.

28. Risks and mitigations

The two largest risks are outside the code: access to a screening provider's API and agreement on the exact thesis format rules.

Risk	Likelihood	Impact	Mitigation
No API access to a similarity / AI screening provider	Medium	High	Build the `ScreeningProvider` interface and the manual-entry path first (FR-THS-09); confirm licensing in sprint 0

Risk	Likelihood	Impact	Mitigation
AI-content scores are unreliable or disputed	High	Medium	Admin override with written justification (FR-THS-10); show the provider's full report; keep the limit configurable
Thesis format rules not agreed in time	Medium	Medium	Ship the default rule set as configurable data; get the official guideline signed off by sprint 5
Format checking of PDFs is limited	High	Low	Prefer DOCX uploads; report which checks were skipped on PDF
Wrong expulsion flag due to bad data (e.g. missing extension)	Medium	High	Flag only, never auto-expel; admin review with full history; extensions recorded in the system
Low adoption by lecturers for attendance	Medium	Medium	One-tap roster with "Mark all present"; optional QR check-in; eligibility lists only from system data
Legacy student data incomplete	High	Medium	CSV import with row-level validation; data cleanup sprint before go-live
Scope creep (examiners, SMS, SSO)	Medium	Medium	Out-of-scope list in section 3; changes go through a change request against this PRD
Email deliverability	Medium	Medium	Use the university SMTP relay or a transactional email service with SPF / DKIM; in-app notifications are the system of record

29. Assumptions and open questions

Development can start now: each item below is either a setting or isolated in one module, so a different answer changes configuration, not architecture.

Assumptions made in this PRD

- The 18-month checkpoint requires 4 approved presentations for both Masters and PhD.
- Presentations can start from semester 2.

- The 20% limit applies separately to the similarity index and to the AI score; exceeding either rejects the thesis.
- Evidence is mandatory for Masters and optional for PhD.
- Expulsion is flagged by the system and confirmed by the admin, never automatic.
- A failed presentation can be re-attempted up to 3 times.
- Attendance threshold (70%) is applied per course; LATE counts as attended and EXCUSED sessions are excluded from the total.
- One primary supervisor per student; co-supervisors are optional.
- The system is hosted by the university (on-premise or its chosen cloud).

Open questions

- [ ] Confirm the 18-month checkpoint count (4) for both programmes.
- [ ] Should similarity and AI score have different limits?
- [ ] Provide the official thesis formatting guideline to finalise the rule set.
- [ ] Which similarity / AI screening provider is licensed, and is API access included?
- [ ] Is there a minimum number of approved publications before thesis submission (Masters and PhD)?
- [ ] Should PhD students also be required to upload evidence?
- [ ] Maximum number of attempts per presentation — is 3 correct?
- [ ] Which frontend framework does the team prefer (React or Angular)?
- [ ] Hosting: on-premise servers or cloud, and who operates them?